

Volatility Malware Memory Analysis — Challenge Write-up

A practical walkthrough for identifying the suspicious process, its executable path, network activity, geographic association, file hash, compilation timestamp, and related domain.

1. Suspicious Process

Question: What is the name of the process responsible for the suspicious activity?

I started by examining the process tree:

```
vol -f file.dmp windows.pstree
```

The process tree showed that userinit.exe launched explorer.exe, which was associated with three processes: OneDrive.exe, SecurityHealth, and ChromeSetup.exe.

ChromeSetup.exe was identified as the suspicious process because the executable was located in the user's Downloads directory and it also attempted to establish a network connection to 58.64.204.181:5202. This behavior made it suspicious compared with the other processes.

userinit.exe is involved in launching processes during user logon, which explains why it appears in this process chain.

Answer: ChromeSetup.exe

2. Exact Executable Path

Question: What is the exact path of the executable for the malicious process?

I used the process tree and filtered for ChromeSetup.exe:

```
vol -f memory.dmp windows.pstree | grep "ChromeSetup.exe"
```

```
# If the path is not visible, try:
vol -f memory.dmp windows.cmdline
```

The executable path appeared as:

```
C:\Users\alex\Downloads\ChromeSetup.exe
```

Answer: C:\Users\alex\Downloads\ChromeSetup.exe

3. Destination IP Address

Question: What IP address did the malware attempt to connect to?

I examined network connections using:

```
vol -f memory.dmp windows.netstat.NetStat
```

The relevant connection was:

```
0xca82b8bc2b30  TCPv4  192.168.19.133  49682  58.64.204.181  5202  SYN_SENT  4628  ChromeSetup.exe  2024
```

This shows that ChromeSetup.exe attempted to establish a TCP connection to 58.64.204.181 on port 5202. The SYN_SENT state indicates that the connection attempt was in progress; it does not by itself prove that the connection was successfully established.

Answer: 58.64.204.181

4. IP Geolocation

Question: Which city is associated with the IP address the malware communicated with?

I took the identified IP address and checked it using an IP geolocation/lookup website to determine the associated city.

Answer: Hong Kong

5. SHA1 Hash of the Malware

Question: What is the SHA1 hash of the malware executable?

First, I located the file object associated with ChromeSetup.exe:

```
vol -f memory.dmp windows.dumpfiles.DumpFiles | grep "ChromeSetup.exe"
```

I then copied the relevant virtual address and dumped the file:

```
vol -f memory.dmp windows.dumpfiles.DumpFiles --virtaddr <file_address>
```

After extracting the executable image, I calculated its SHA1 hash using:

```
sha1sum <file>.img
```

Volatility returned multiple dumped representations of the file. I used the .img output because, in this challenge, it contained the executable image required for the hash calculation.

Answer: SHA1 hash of the extracted .img executable dump

6. Compilation Timestamp

Question: What is the compilation timestamp for the malware?

I used VirusTotal to examine the malware sample and its metadata/history. The compilation timestamp identified for the malware was:

Answer: 2019-12-01 08:36

Note: PE metadata can also be examined locally with tools such as exiftool. Different timestamps may be present in a sample, so the timestamp should be verified as the PE compilation timestamp rather than a submission, modification, or analysis timestamp.

7. Domain Associated with the Malware

Question: Can you provide the domain connected to the malware?

VirusTotal showed the hostname/subdomain:

```
ddos.dnsnb8.net
```

Because the challenge asks for the main/base domain, the subdomain was reduced to:

Answer: dnsnb8.net